

Restaurant Recommendation System

A Project Report Submitted
to Smart-Internz for the
completion of
Applied Data Science Internship Program Prepared

By

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Academic Year: 2025–2026

Submitted To

Smart-Internz
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1. Introduction

The Smart Restaurant Recommendation System is a web-based application designed to help users find restaurants based on their preferences such as cuisine, budget, and ratings.

It uses machine learning techniques to analyze restaurant data and provide personalized recommendations.

1.1 Project Overview

This system recommends restaurants similar to a selected restaurant using **Content-Based Filtering**.

It processes restaurant data, calculates similarity between restaurants, and displays relevant results through a user-friendly interface built using Flask.

2. Objectives

- To build a recommendation system using Content-Based Filtering
 - To analyze and preprocess restaurant dataset
 - To apply machine learning algorithms (TF-IDF and Cosine Similarity)
 - To develop a web-based application using Flask
 - To provide personalized restaurant suggestions
-

3. Project Initialization and Planning Phase

3.1 Define Problem Statement

Users often find it difficult to choose restaurants based on their preferences due to a large number of options available online.

The problem is to build a system that provides personalized restaurant recommendations based on user input.

3.2 Project Proposal (Proposed Solution)

The proposed solution is to develop a recommendation system that:

- Takes user input (restaurant name, cuisine, budget, rating)
 - Uses machine learning techniques to find similar restaurants
 - Displays filtered results based on preferences
-

3.3 Initial Project Planning

- Select dataset
 - Choose algorithm (Content-Based Filtering)
 - Design system architecture
 - Develop backend using Flask
 - Design frontend UI
 - Test and deploy system
-

4. Data Collection and Preprocessing Phase

4.1 Data Collection Plan and Raw Data Sources Identified

The dataset used contains restaurant details such as:

- Name
- Rating
- Mean Rating
- Cuisines
- Cost

The dataset is collected in CSV format.

4.2 Data Quality Report

- Some entries had missing values
- Duplicate restaurant names existed
- Inconsistent text formatting (uppercase/lowercase) These

issues were handled during preprocessing.

4.3 Data Preprocessing

- Converted text to lowercase
 - Removed duplicates
 - Handled missing values
 - Cleaned cuisine data
 - Reset index for proper data handling
-

5. Model Development Phase

5.1 Model Selection Report

Content-Based Filtering was selected because:

- It does not require user history
 - Works well with textual data (cuisines)
 - Easy to implement and interpret
-

5.2 Initial Model Training, Validation and Evaluation Report

- TF-IDF Vectorizer was used to convert cuisine text into numerical vectors
- Cosine Similarity was used to measure similarity between restaurants

The model was tested by checking whether similar restaurants were correctly recommended.

6. Model Optimization and Tuning Phase

6.1 Tuning Documentation

- Improved dataset by adding more cuisine keywords
 - Adjusted filtering conditions (budget, rating)
 - Removed duplicate recommendations
 - Fixed indexing issues
-

6.2 Final Model Selection Justification

The final model uses:

- TF-IDF Vectorization
- Cosine Similarity
- Post-filtering (cuisine, budget, rating)

This combination provides accurate and relevant recommendations.

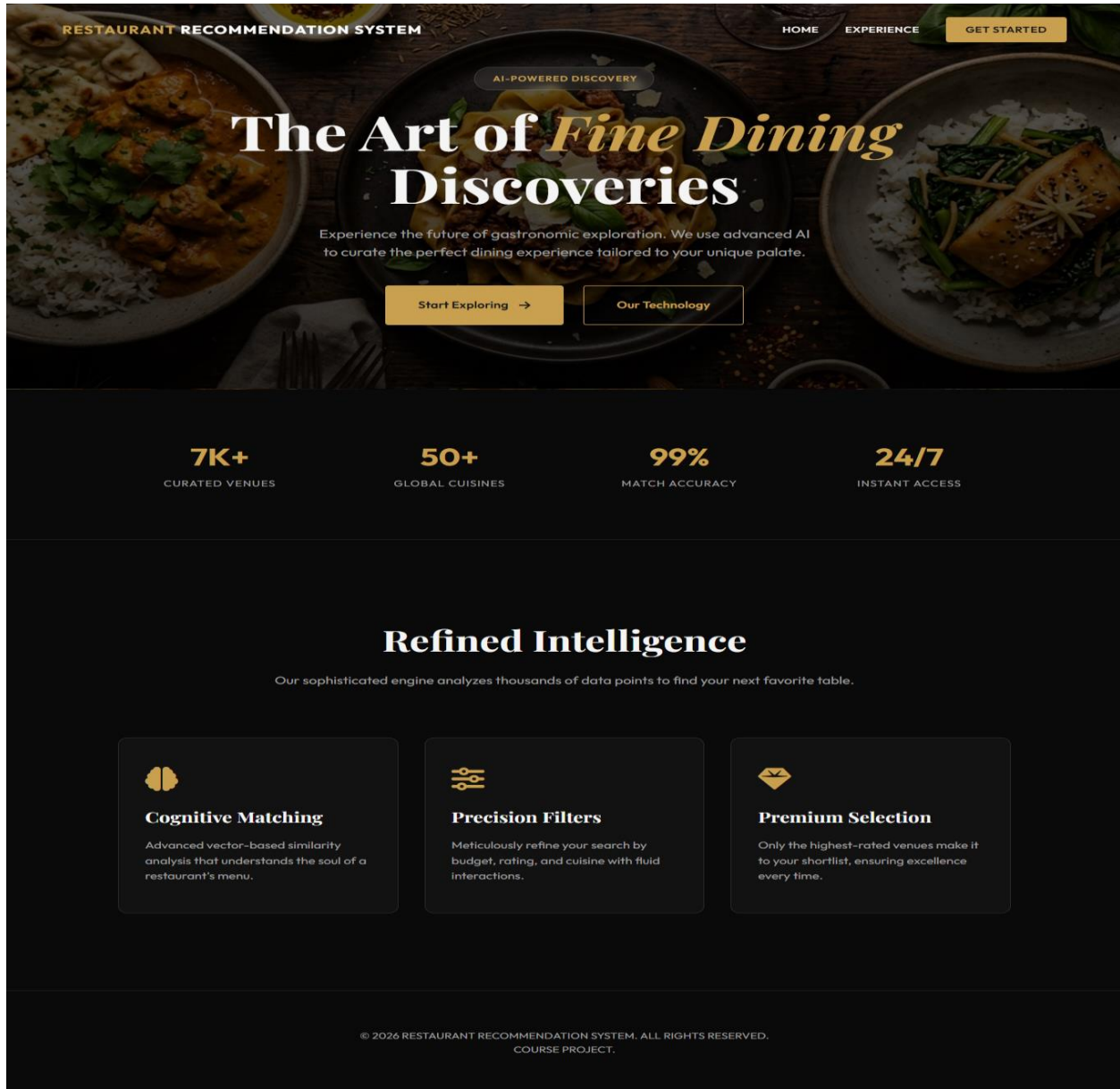
7. Results

The system successfully recommends restaurants based on user input.

- Similar restaurants are displayed
- Filters are applied correctly
- UI displays results in card format

7.1 Output Screenshots

- Home Page



- Recommendation Form

RESTAURANT RECOMMENDATION SYSTEM

HOMEEXPLORE

Search Venues

Inspiration Restaurant

Enter a restaurant you love...

Preferred Cuisine

Any Cuisine

Maximum Budget (₹)

No Limit

Minimum Rating

Any Rating

Find Similar

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- Result Page

RESTAURANT RECOMMENDATION SYSTEM

HOMENEW SEARCH

← Back to search

Results for Cafe Coffee Day

We've found 3 venues that match your refined taste.



Tummy Fuel

Cafe Beverages Fast Food

★ 4.0 📍 Bengaluru

₹400 / person

View Details



Rooftop Cafe

Cafe Fast Food Beverages

★ 3.8 📍 Bengaluru

₹400 / person

View Details



Cross Road Cafe

Cafe Fast Food Chinese

★ 3.5 📍 Bengaluru

₹300 / person

View Details

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8. Advantages and Disadvantages

8.1 Advantages

- Provides personalized recommendations
 - Fast and efficient system
 - Easy to use interface
 - No need for user history
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8.2 Disadvantages

- Depends on dataset quality
 - Limited to available data
 - Does not consider user reviews or real-time data
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9. Conclusion

The Smart Restaurant Recommendation System successfully demonstrates how machine learning can be used to provide personalized suggestions.

It improves user experience and helps users make better decisions.

10. Future Scope

- Add location-based recommendations
 - Integrate real-time APIs (Zomato/Swiggy)
 - Add user login and preferences
 - Use collaborative filtering
 - Improve UI with images and reviews
-

11. Appendix

11.2 GitHub & Project Video Demo Link

Add:

- GitHub Repository Link

<https://github.com/Atharvasawant10/restaurant-recommendation-system>

- Project Demo Video Link :

<https://drive.google.com/file/d/1YDBXxbh68gyY6R9VzNXtua93STwUkSFI/view?usp=sharing>